Digital System Design

Textbook : Malvino P Leech
Reference : Morris Mano, Sudhakar Samuel

Boolean Algebra

Variables			Min term		Max term	
x	y	z	Term	Designation	Term	Designation
0	0	0	$x'y'z'$	m_0	$(x+y+z)$	M_0
0	0	1	$x'y'z$	m_1	$(x+y+z')$	M_1
0	1	0	$x'yz'$	m_2	$(x+y'+z)$	M_2
0	1	1	$x'yz$	m_3	$(x+y'+z')$	M_3
1	0	0	$xy'z'$	m_4	$(x'+y+z)$	M_4
1	0	1	$xy'z$	m_5	$(x'+y+z')$	M_5
1	1	0	xyz'	m_6	$(x'+y'+z)$	M_6
1	1	1	xyz	m_7	$(x'+y'+z')$	M_7

Q: $F(A,B,C) = AB + BC$. Express in minterm format.

Ans: $F(A,B,C) = AB + BC$

$= AB(C+\bar{C}) + (A+\bar{A})BC$

$= ABC + AB\bar{C} + ABC + \bar{A}BC$

$= ABC + AB\bar{C} + \bar{A}BC$

$= m_7 + m_6 + m_3$

$= \Sigma_m(3,6,7)$

Q: $f(A,B,C,D) = BC + AD$

Ans: $f(A,B,C,D) = (A+\bar{A})BC(D+\bar{D}) + A(B+\bar{B})(C+\bar{C})D$

$= ABC(D+\bar{D}) + \bar{A}BC(D+\bar{D}) + AB(C+\bar{C})D + A\bar{B}(C+\bar{C})D$

$= ABCD + ABC\bar{D} + \bar{A}BCD + \bar{A}BC\bar{D} + ABCD + AB\bar{C}D$

$+ A\bar{B}CD + A\bar{B}\bar{C}D$

$= m_{15} + m_{14} + m_7 + m_6 + m_{15} + m_{13} + m_{11} + m_9$

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$= \Sigma_m(6, 7, 9, 11, 13, 14, 15)$

Q: $f(A,B,C) = (A+B)(A+C)$

$= A + AB + AC + BC$

$= A(B+\bar{B})(C+\bar{C}) + AB(C+\bar{C}) + A(B+\bar{B})C + (A+A)BC$

$= AB(C+\bar{C}) + A\bar{B}(C+\bar{C}) + ABC + AB\bar{C} + ABC + A\bar{B}C$

$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC + AB\bar{C} + ABC + \bar{A}BC$

$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}BC$

$= m_7 + m_6 + m_5 + m_4 + m_3$

$= \Sigma_m(3,4,5,6,7)$

Q: Write the truth table for following.

$f = \pi_m(2,7)$

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Q: Write algebraic form of boolean expression $f(A,B,C,D) = \Sigma_m(3,4,9,12)$

Ans: $f(A,B,C,D) = \bar{A}\bar{B}CD + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}D + AB\bar{C}\bar{D}$

Q: Represent the following in both max and min term form. $F(X,Y,Z) = \bar{X}Y + YZ$

Ans: $F(X,Y,Z) = \bar{X}Y(Z+\bar{Z}) + YZ$

$= \bar{X}YZ + \bar{X}Y\bar{Z} + XYZ + \bar{X}YZ$

$= m_3 + m_2 + m_7 = \Sigma_m(2,3,7) = \pi_M(0,1,4,5,6)$

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Q: Simplify the boolean expression using 3 variable K map and implement result using basic gates: $F(x,y,z) = xy + x'y'z' + x'yz'$

Ans: $F(x,y,z) = xy(z+z') + x'y'z' + x'yz'$

$= xyz + xyz' + x'y'z' + x'yz'$

$= m_7 + m_6 + m_0 + m_2$

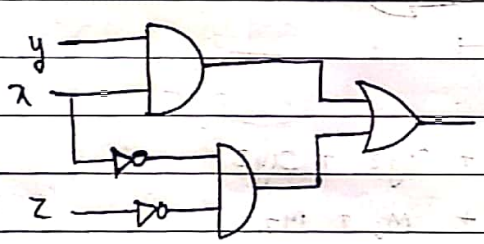
$= \Sigma_m(0,2,6,7)$

x \ yz	y'z'	y'z	yz	yz'
x'	1 (0)	0 (1)	0 (3)	1 (2)
x	0 (4)	0 (5)	1 (7)	1 (6)

$x'y'z' + x'yz' + xyz' + xyz$

$= x'z'(y'+y) + xy(z'+z)$

$= x'z' + xy$



Q: $F(x,y,z) = x'y' + yz + x'yz$

Ans: $F(x,y,z) = x'y'(z+z') + (x+x')yz + x'yz$

$= x'y'z + x'y'z' + xyz + x'yz + x'yz$

$= m_1 + m_0 + m_7 + m_3$

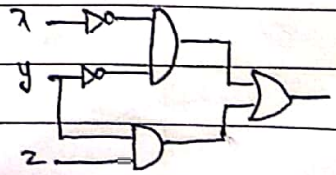
$= \Sigma_m(0,1,3,7)$

x \ yz	y'z'	y'z	yz	yz'
x'	1 (0)	1 (1)	1 (3)	0 (2)
x	0 (4)	0 (5)	1 (7)	0 (6)

$x'y'z' + x'y'z + x'yz + xyz$

$= x'y'(z'+z) + yz(x+x')$

$= x'y' + yz$



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Q: $F(x,y,z) = x'y + yz' + y'z'$

$= x'y(z+z') + (x+x')yz' + (x+x')y'z'$

$= x'yz + x'yz' + xyz' + x'yz' + xy'z' + x'y'z'$

$= m_3 + m_2 + m_6 + m_4 + m_0$

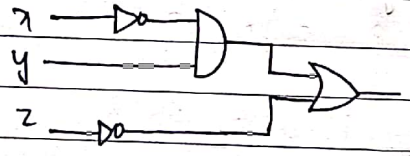
$= \Sigma_m(0, 2, 3, 4, 6)$

	$y'z'$	$y'z$	yz	yz'
x'	1 (0)	0 (1)	1 (3)	1 (2)
x	1 (4)	0 (5)	0 (7)	1 (6)

$x'y'z' + xy'z' + x'yz' + xyz' + x'yz + x'yz'$

$= z'(x'y' + xy + xy' + x'y) + x'y(z+z')$

$= z' + x'y$



Q: $F(x,y,z) = x'y'z + xyz' + xyz$

$= m_1 + m_6 + m_7$

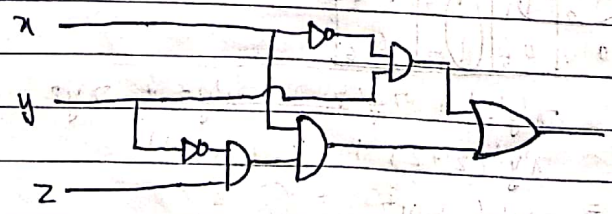
$= \Sigma_m(1, 6, 7)$

	$y'z'$	$y'z$	yz	yz'
x	0 (0)	1 (1)	0 (3)	0 (2)
x'	0 (4)	0 (5)	1 (7)	1 (6)

$x'yz + x'yz' + xy'z$

$= x'y(z+z') + xy'z$

$= x'y + xy'z$



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Q: Simplify boolean function - $F(x,y,z) = \Sigma_m(3,5,6,7)$

	$y'z'$	$y'z$	yz	yz'
x'	0 0	0 1	1 3	0 2
x	0 4	1 5	1 7	1 6

$xz + xy + yz$

Q: Simplify the boolean function using 4 variable KC map.

$F(A,B,C,D) = \Sigma_m(3,7,11,13,14,15)$

	$C'D'$	$C'D$	CD	CD'
$A'B'$	0 0	0 1	1 3	0 2
$A'B$	0 4	0 5	1 7	0 6
AB	0 12	1 13	1 15	1 14
AB'	0 8	0 9	1 11	0 10

$CD + ABD + ABC$

Q: $F(A,B,C,D) = \Sigma_m(4,6,7,15)$

	$C'D'$	$C'D$	CD	CD'
$A'B'$	0 0	0 1	0 3	0 2
$A'B$	1 4	0 5	1 7	1 6
AB	0 12	0 13	1 15	0 14
AB'	0 8	0 9	0 11	0 10

$BCD + A'BD'$

Q: $F(A,B,C,D) = \Sigma_m(0,1,4,5,6,7,8,9)$

	$C'D'$	$C'D$	CD	CD'
$A'B'$	1 0	1 1	0 3	0 2
$A'B$	1 4	1 5	1 7	1 6
AB	0 12	0 13	0 15	0 14
AB'	1 8	1 9	0 11	0 10

$A'B + B'C'$

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Q: $F(W,X,Y,Z) = \Sigma_m(0,1,2,4,5,6,8,9,12,13,14)$

	$Y'Z'$	$Y'Z$	YZ	YZ'
$W'X'$	1 (0)	1 (1)	0 (3)	1 (2)
$W'X$	1 (4)	1 (5)	0 (7)	1 (6)
WX	1 (12)	1 (13)	0 (15)	1 (14)
WX'	1 (8)	1 (9)	0 (11)	0 (10)

$Y' + XZ' + W'Z'$

Q: $F(A,B,C,D) = A'B'C' + B'CD' + A'BCD' + AB'C'$

$= A'B'C'(D+D') + (A+A')B'CD' + A'BCD' + AB'C'(D+D')$

$= A'B'C'D + A'B'C'D' + AB'CD' + A'B'CD' + A'BCD' + AB'C'D + AB'C'D'$

$= m_1 + m_0 + m_{10} + m_2 + m_6 + m_9 + m_8$

$= \Sigma_m(0,1,2,6,8,9,10)$

	$C'D'$	$C'D$	CD	CD'
$A'B'$	1 (0)	1 (1)	0 (3)	1 (2)
$A'B$	0 (4)	0 (5)	0 (7)	1 (6)
AB	0 (12)	0 (13)	0 (15)	0 (14)
AB'	1 (8)	1 (9)	0 (11)	1 (10)

$B'C' + B'D' + A'CD'$

Q: $F(A,B,C,D) = \Sigma_m(0,2,3,5,7,8,9,10,11,13,15)$

	$C'D'$	$C'D$	CD	CD'
$A'B'$	1 (0)	0 (1)	1 (3)	1 (2)
$A'B$	0 (4)	1 (5)	1 (7)	0 (6)
AB	0 (12)	1 (13)	1 (15)	0 (14)
AB'	1 (8)	1 (9)	1 (11)	1 (10)

$B'D' + BD + CD + AB'$

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Q: Consider the functions, $F_1(A,B,C,D) = \Sigma_m(1,3,4,12,13)$

$F_2(A,B,C,D) = \Sigma_m(0,1,3,4,6,8,10,11,12,13)$

Find F_1+F_2, $F_1 \cdot F_2$

$F_1+F_2 = F_2$

	C'D'	C'D	CD	CD'
A'B'	1 0	1 1	1 3	0 2
A'B	1 4	0 5	0 7	1 6
AB	1 12	1 13	0 15	0 14
AB'	1 8	0 9	1 11	1 10

$C'D' + A'B'D + A'BD' + ABC' + AB'C$

$F_1 \cdot F_2 = F_1$

	C'D'	C'D	CD	CD'
A'B'	0 0	1 1	1 3	0 2
A'B	1 4	0 5	0 7	0 6
AB	1 12	1 13	0 15	0 14
AB'	0 8	0 9	0 11	0 10

$A'B'D + BC'D' + ABC'$

Q: Simplify in SoP and PoS. $F(A,B,C,D) = \Sigma_m(0,1,2,5,8,9,10)$

	C'D'	C'D	CD	CD'
A'B'	1 0	1 1	0 3	0 2
A'B	0 4	1 5	0 7	0 6
AB	0 12	0 13	0 15	0 14
AB'	1 8	1 9	0 11	1 10

$B'D' + B'C' + A'C'D$

	C+D	C+D'	C'+D'	C'+D
A+B	1 0	1 1	0 3	1 2
A+B'	0 4	1 5	0 7	0 6
A'+B'	0 12	0 13	0 15	0 14
A'+B	1 8	1 9	0 11	1 10

$(A'+B').(C'+D').(B'+D)$

Q: Verify the statement by simplifying K-map. $F(A,B,C,D) = A + \bar{C}$

$F(A,B,C,D) = A(B+\bar{B})(C+\bar{C})(D+\bar{D}) + (A+\bar{A})(B+\bar{B})\bar{C}(D+\bar{D})$

$= ABCD + ABC\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D}$
$+ A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}\bar{C}D$
$+ A\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$

$= m_{15} + m_{14} + m_{13} + m_{12} + m_{11} + m_{10} + m_9 + m_8$
$+ m_5 + m_4 + m_1 + m_0$

$= \Sigma_m(0,1,4,5,8,9,10,11,12,13,14,15)$

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Q: Consider the functions, $F_1(A,B,C,D) = \Sigma_m(1,3,4,12,13)$

$F_2(A,B,C,D) = \Sigma_m(0,1,3,4,6,8,10,11,12,13)$

Find F_1+F_2, $F_1 \cdot F_2$

$F_1+F_2 = F_2$

	C'D'	C'D	CD	CD'
A'B'	1 0	1 1	1 3	0 2
A'B	1 4	0 5	0 7	1 6
AB	1 12	1 13	0 15	0 14
AB'	1 8	0 9	1 11	1 10

$C'D' + A'B'D + A'BD' + ABC' + AB'C$

$F_1 \cdot F_2 = F_1$

	C'D'	C'D	CD	CD'
A'B'	0 0	1 1	1 3	0 2
A'B	1 4	0 5	0 7	0 6
AB	1 12	1 13	0 15	0 14
AB'	0 8	0 9	0 11	0 10

$A'B'D + BC'D' + ABC'$

Q: Simplify in SoP and PoS - $F(A,B,C,D) = \Sigma_m(0,1,2,5,8,9,10)$

	C'D'	C'D	CD	CD'
A'B'	1 0	1 1	0 3	0 2
A'B	0 4	1 5	0 7	0 6
AB	0 12	0 13	0 15	0 14
AB'	1 8	1 9	0 11	1 10

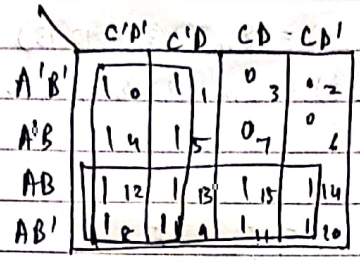
$B'D' + B'C' + A'C'D$

	C+D	C+D'	C'+D'	C'+D
A+B	1 0	1 1	0 3	1 2
A+B'	0 4	1 5	0 7	0 6
A'+B'	0 12	0 13	0 15	0 14
A'+B	1 8	1 9	0 11	1 10

$(A'+B').(C'+D').(B'+D)$

Q: Verify the statement by simplifying K-map. $F(A,B,C,D) = A + \bar{C}$

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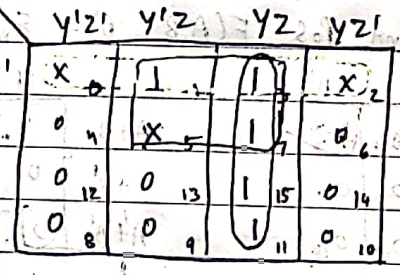


$A + \bar{C}$

SoP → Prime implicant
PoS → Prime implicate

Q: Simplify using K-map. $F(w,x,y,z) = \Sigma_m(1,3,7,11,15)$

$d(w,x,y,z) = \Sigma_d(0,2,5)$

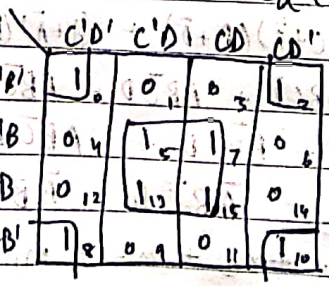


$yz + w'z$

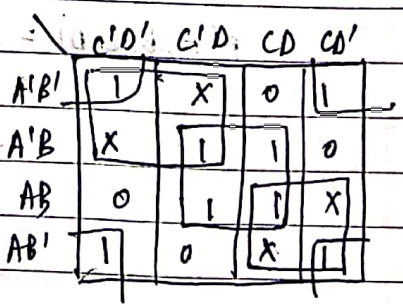
Q: Identify prime implicants and essential prime implicants.

$f(A,B,C,D) = \Sigma_m(0,2,5,7,8,10,13,15)$

$\Sigma_d(1,4,11,14)$



$B'D' + BD$



$B'D' + BD + A'C' + AC$

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Q: $F(A,B,C,D) = \Pi_M(0,1,4,5,8,9,11) + d(2,10)$. Identify prime implicates.

	C+D	C+D'	C'+D'	C'+D
A+B	0 (0)	0 (1)	1 (3)	X (2)
A+B'	0 (4)	0 (5)	1 (7)	1 (6)
A'+B'	1 (12)	1 (13)	1 (15)	1 (14)
A'+B	0 (8)	0 (9)	0 (11)	X (10)

$(A+C)(A'+B)$

5 variable K-map

	000	001	011	010	110	111	101	100
00	0	1	3	2	6	7	5	4
01	8	9	11	10	14	15	13	12
11	24	25	27	26	30	31	29	28
10	16	17	19	18	22	23	21	20

Q: Simplify the following function: $F(A,B,C,D,E) = \Sigma_m(5,7,13,15,21,23,29,31)$

	$\bar{C}\bar{D}\bar{E}$	$\bar{C}\bar{D}E$	$\bar{C}DE$	$\bar{C}D\bar{E}$	$CD\bar{E}$	CDE	$C\bar{D}E$	$C\bar{D}\bar{E}$
$\bar{A}\bar{B}$	0	0	0	0	0	1	1	0
$\bar{A}B$	0	0	0	0	0	1	1	0
AB	0	0	0	0	0	1	1	0
$A\bar{B}$	0	0	0	0	0	1	1	0

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Q: $F(A,B,C,D,E) = \Sigma_m(0,2,4,6,9,13,21,23,25,29,31)$

	$\bar{C}\bar{D}\bar{E}$	$\bar{C}\bar{D}E$	$\bar{C}DE$	$\bar{C}D\bar{E}$	$CD\bar{E}$	CDE	$C\bar{D}E$	$C\bar{D}\bar{E}$
$\bar{A}\bar{B}$	1	0	0	1	1	0	0	1
$\bar{A}B$	0	1	0	0	0	0	1	0
AB	0	1	0	0	0	1	1	0
$A\bar{B}$	0	0	0	0	0	1	1	0

$ACE + \bar{A}\bar{B}D\bar{E} + B\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{D}\bar{E} + BC\bar{D}E$

$= ACE + \bar{A}\bar{B}\bar{E} + B\bar{D}E$

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Q: $F(A,B,C,D,E) = \Sigma_m(0,4,5,8,12,13,16,20,21,24,28,29)$

	$\bar{C}\bar{D}\bar{E}$	$\bar{C}\bar{D}E$	$\bar{C}D\bar{E}$	$\bar{C}DE$	$C\bar{D}\bar{E}$	$C\bar{D}E$	$CD\bar{E}$	CDE
$\bar{A}\bar{B}$	1 0	0 1	0 3	0 2	0 6	0 7	1 5	1 4
$\bar{A}B$	1 8	0 9	0 11	0 10	0 14	0 15	1 13	1 12
AB	1 24	0 25	0 27	0 26	0 30	0 31	1 29	1 28
$A\bar{B}$	1 16	0 17	0 19	0 18	0 22	0 23	1 21	1 20

$C\bar{D} + \bar{D}\bar{E}$

Q: $f(a,b,c,d,e,f) = \Sigma_m(0,2,8,10,16,18,24,26,32,34,40,42,48,50,56,58)$

	$\bar{D}\bar{E}\bar{F}$	$\bar{D}\bar{E}F$	$\bar{D}EF$	$\bar{D}E\bar{F}$	$DE\bar{F}$	DEF	$D\bar{E}F$	$D\bar{E}\bar{F}$
$\bar{A}\bar{B}\bar{C}$	1 0	0 1	0 3	1 2	0 6	0 7	0 5	0 4
$\bar{A}\bar{B}C$	1 8	0 9	0 11	1 10	0 14	0 15	0 13	0 12
$\bar{A}BC$	1 24	0 25	0 27	1 26	0 30	0 31	0 29	0 28
$\bar{A}B\bar{C}$	1 16	0 17	0 19	1 18	0 22	0 23	0 21	0 20
$A\bar{B}\bar{C}$	1 48	0 49	0 51	1 50	0 54	0 55	0 53	0 52
ABC	1 56	0 57	0 59	1 58	0 62	0 63	0 61	0 60
$A\bar{B}C$	1 40	0 41	0 43	1 42	0 46	0 47	0 45	0 44
$A\bar{B}\bar{C}$	1 32	0 33	0 35	1 34	0 38	0 39	0 37	0 36

$\bar{D}\bar{E}\bar{F} + \bar{D}E\bar{F} = \bar{D}\bar{F}(\bar{E}+E)$

$= \bar{D}\bar{F}$

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Quine McClusky (Tabulation Method)

Q: Obtain all the prime implicants of $f(A,B,C,D) = \Sigma_m(0,2,3,5,8,10,11)$

Ans:

Index	A B C D			
0	0 0 0 0	✓	(0,2) 0 0 - 0 ✓	(0,2,8,10) - 0 - 0
2	0 0 1 0	✓	(0,8) - 0 0 0 ✓	(0,8,2,10) - 0 - 0
8	1 0 0 0	✓	(2,3) 0 0 1 - ✓	(2,3,10,11) - 01 -
3	0 0 1 1	✓	(2,10) - 0 1 0 ✓	(2,10,3,11) - 01 -
5	0 1 0 1		(8,10) 1 0 - 0 ✓	
10	1 0 1 0	✓	(3,11) - 0 1 1 ✓	
11	1 0 1 1	✓	(10,11) 1 0 1 - ✓	

$$B'D' + B'C + A'BC'D$$

Q: Solve and find prime implicants. $f(AB,C,D) = \Sigma_m(4,5,7,12,14,15)$

Ans:

	A B C D			
4	0 1 0 0	✓	(4,5) 0 1 0 - *	
5	0 1 0 1	✓	(4,12) - 1 0 0 *	
12	1 1 0 0	✓	(5,7) 0 1 - 1 *	
7	0 1 1 1	✓	(12,14) 1 1 - 0 *	
14	1 1 1 0	✓	(7,15) - 1 1 1 *	
15	1 1 1 1	✓	(14,15) 1 1 1 - *	

$$\bar{A}B\bar{C} + B\bar{C}\bar{D} + \bar{A}BD + AB\bar{D} + BCD + ABC$$

Q: $f(a,b,c,d) = \pi_M(0,6,7,8,9,13) + \pi_d(5,15)$

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	A B C D			
0	0 0 0 0 ✓	(0,8) - 0 0 0 *	(5,13,7,15) - 1 - 1 *
8	1 0 0 0 ✓	(8,9) 1 0 0 - *	(5,7,13,15) - 1 - 1 *
5	0 1 0 1 ✓	(5,7) 0 1 - 1 *	
6	0 1 1 0 ✓	(6,7) 0 1 1 - *	
9	1 0 0 1 ✓	(5,13) - 1 0 1 ✓	
7	0 1 1 1 ✓	(9,13) 1 - 0 1 *	
13	1 1 0 1 ✓	(7,15) - 1 1 1 ✓	
15	1 1 1 1 ✓	(13,15) 1 1 - 1 *	

$\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C} + \bar{A}BC + A\bar{C}D + BD$

$(B+C+D)(\bar{A}+B+C)(A+\bar{B}+\bar{C})(\bar{A}+C+\bar{D})(\bar{B}+\bar{D})$ (Prime implicate)

	0	6	7	8	9	13
(0,8) ✓	x			x		
(8,9)				x	x	
(6,7) ✓		x	x			
(9,13)					x	x
(5,7,13,15)			x			x
	✓	✓	x	x	x	x

$(B+C+D)(A+\bar{B}+\bar{C})$ (Essential prime implicates)

Q: Apply Petrick's method to find essential prime implicants.

$f(a,b,c) = \Sigma_m(2,3,4,5,7)$

Ans:

	A B C		
2	0 1 0 ✓	(2,3) 0 1 - *	
4	1 0 0 ✓	(4,5) 1 0 - *	
3	0 1 1 ✓	(3,7) - 1 1 *	
5	1 0 1 ✓	(5,7) 1 - 1 *	
7	1 1 1 ✓		

$\bar{A}B + A\bar{B} + BC + AC$

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W	(2,3)	X	X		
X	(4,5)			X	X
Y	(3,7)		X		X
Z	(5,7)			X	X

$W.(W+Y).X.(X+Z).(Y+Z)$

$(WW+WY)(XX+XZ)(Y+Z)$

$(W+WY)(X+XZ)(Y+Z)$

$W.X.(Y+Z)$

$WXY + WXZ$

$\bar{A}B + A\bar{B} + BC$ or $\bar{A}B + A\bar{B} + AC$

Q: Identify essential prime implicants by Petrick method. $\Sigma m(0,1,2,5,6,7)$

Ans:

	A	B	C			
0	0	0	0	✓	(0,1) 00 − *	$\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}C$
1	0	0	1	✓	(0,2) 0 − 0 *	$+ B\bar{C} + AC + AB$
2	0	1	0	✓	(1,5) − 01 *	
5	1	0	1	✓	(2,6) − 10 *	
6	1	1	0	✓	(5,7) 1 − 1 *	
7	1	1	1	✓	(6,7) 11 − *	

		0	1	2	5	6	7
U	(0,1)	X	X				
V	(0,2)	X		X			
W	(1,5)		X		X		
X	(2,6)			X		X	
Y	(5,7)				X		X
Z	(6,7)					X	X

$(U+V)(U+W)(V+X)(W+Y)(X+Z)(Y+Z)$

$(U+UW+UV+VW)(VX+VZ+X+XZ)(WY+WZ+Y+YZ)$

$[U(1+W+V)+VW][X(1+V+Z)+VZ][Y(1+W+Z)+WZ]$

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$(U+VW)(X+VZ)(Y+WZ)$

$= (UX + UVZ + VWX + VWZ)(Y+WZ)$

$= UXY + UVZY + VWXY + VWYZ + UXWZ + UVWZ + VWZ$
$+ VWZ$

$\bar{A}\bar{B} + B\bar{C} + AC$ or $\bar{A}\bar{C} + \bar{B}C + AB$

Combinational Circuits with SSI logic

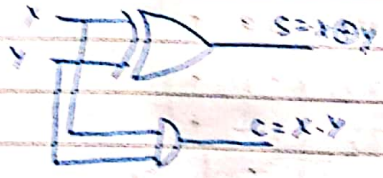
Adders - Half adder, Full adder, Parallel adder

Half adder

x	y	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

	$\bar{y}$	y
$\bar{x}$	0	1
x	1	0

$S = \bar{x}y + x\bar{y} = x \oplus y$



	$\bar{y}$	y
$\bar{x}$	0	0
x	0	1

$C = xy = x.y$

Full adder

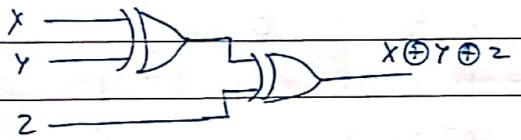
x	y	z	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

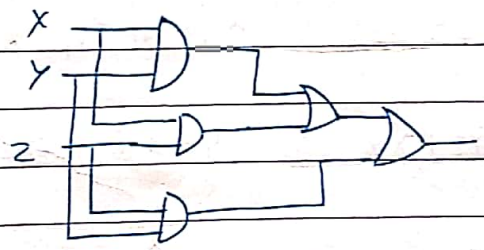
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		$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
S	$\bar{x}$	0_0	1_1	0_3	1_2
	x	1_4	0_5	1_7	0_6
C	$\bar{x}$	0_0	0_1	1_3	0_2
	x	0_4	1_5	1_7	1_6

$$S = x\bar{y}\bar{z} + \bar{x}\bar{y}z + xyz + \bar{x}y\bar{z}$$
$$= x(yz + \bar{y}\bar{z}) + \bar{x}(\bar{y}z + y\bar{z})$$
$$= x(\overline{y \oplus z}) + \bar{x}(y \oplus z)$$
$$= x \oplus y \oplus z$$

$$C = xz + xy + yz$$





Subtractor

Half subtractor

x	y	Borrow	Difference
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

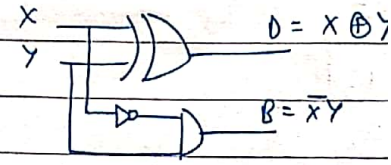
	$\bar{y}$	y
$\bar{x}$	0_0	1_1
x	0_2	0_3

$B = \bar{x}y$

	$\bar{y}$	y
$\bar{x}$	0_0	1_1
x	1_2	0_3

$D = \bar{x}y + x\bar{y}$
$= x \oplus y$

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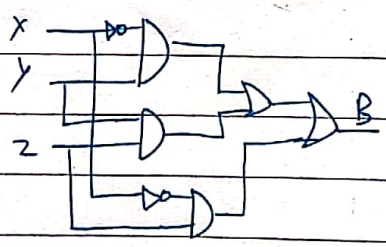


Full Subtractor

x	y	z	Borrow
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	0_0	1_1	1_3	1_2
x	0_4	0_5	1_7	0_6

$$B = \bar{x}z + yz + \bar{x}y$$



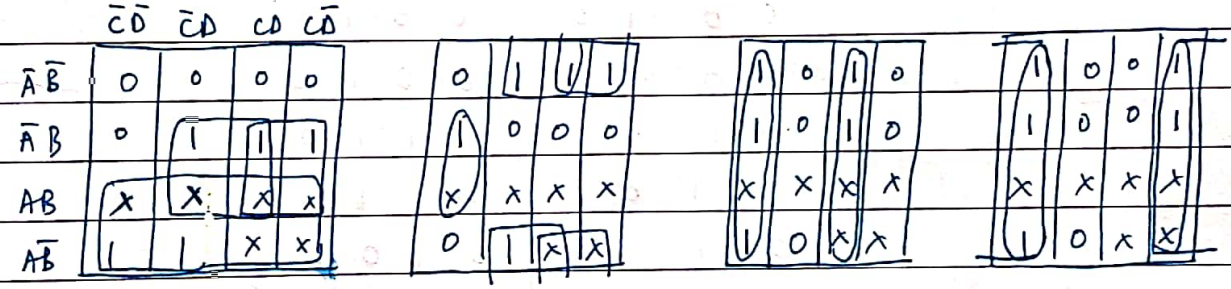
Code Converter

Converting code from one code

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BCD to Excess 3

A	B	C	D	w	x	y	z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0



$A + BD + BC$

$B\bar{C}\bar{D} + \bar{B}D + \bar{B}C$

$\bar{C}\bar{D} + CD$

$= \overline{C \oplus D}$

$\bar{C}\bar{D} + C\bar{D}$

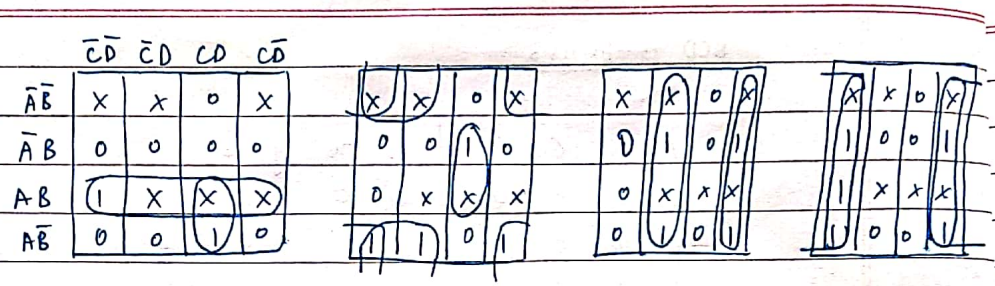
$= \bar{D}(\bar{C} + C)$

$= \bar{D}$

Excess-3 to BCD

A	B	C	D	w	x	y	z
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0
1	0	0	0	0	1	0	1
1	0	0	1	0	1	1	0
1	0	1	0	0	1	1	1
1	0	1	1	1	0	0	0
1	1	0	0	1	0	0	1

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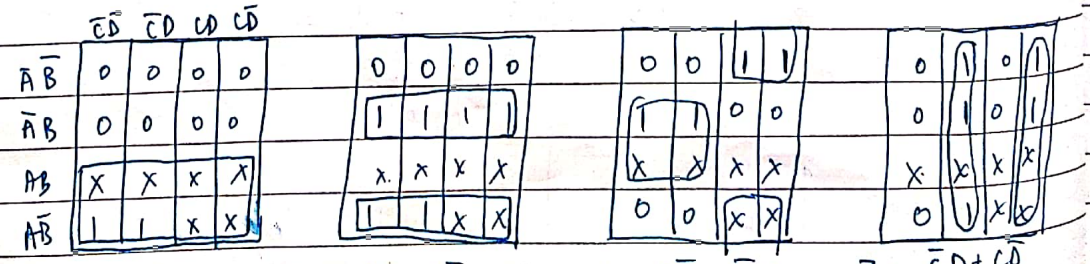


$W = AB + ACD$ $X = \bar{B}\bar{C} + \bar{B}\bar{D} + BCD$ $Y = \bar{C}D + C\bar{D} = C \oplus D$ $Z = \bar{C}\bar{D} + C\bar{D} = \bar{D}$

BCD to Gray Code

A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

Don't Care

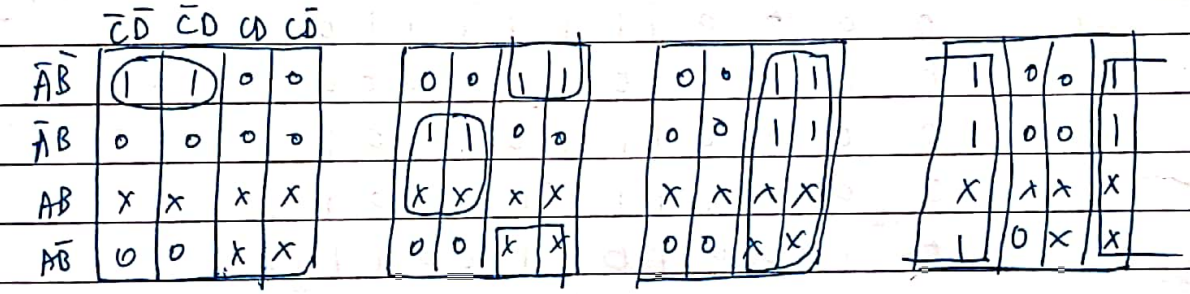


$W = A$ $X = A\bar{B} + \bar{A}B$ $Y = B\bar{C} + \bar{B}C$ $Z = \bar{C}D + C\bar{D}$

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BCD to 9's complement

A	B	C	D	W	X	Y	Z
0	0	0	0	1	0	0	1
0	0	0	1	1	0	0	0
0	0	1	0	0	1	1	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	0	0
0	1	1	0	0	0	1	1
0	1	1	1	0	0	1	0
1	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0



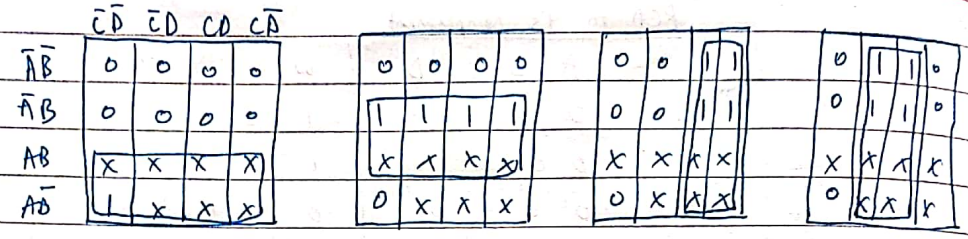
$W = \bar{A}\bar{B}\bar{C}$ $\quad X = B\bar{C} + \bar{B}C$ $\quad Y = \bar{C}$ $\quad Z = \bar{D}$

$= B \oplus C$

8 -4 -2 -1 to BCD

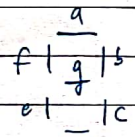
A	B	C	D	w	x	y	z
0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	1
1	1	1	0	0	0	1	0
1	1	0	1	0	0	1	1
1	1	0	0	0	1	0	0
1	0	1	1	0	1	0	1
1	0	1	0	0	1	1	0
1	0	0	1	0	1	1	1
1	0	0	0	1	0	0	0

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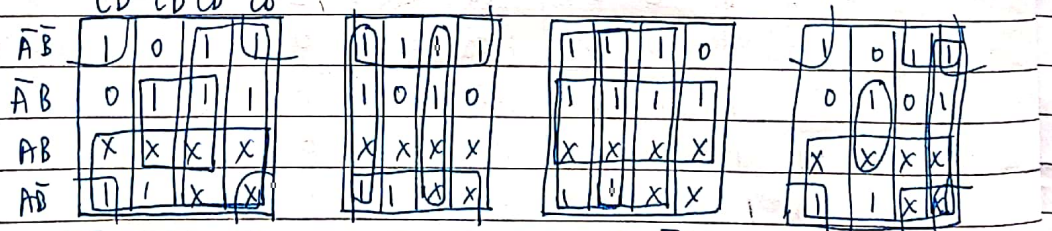


$W = A$ $X = B$ $Y = C$ $Z = D$

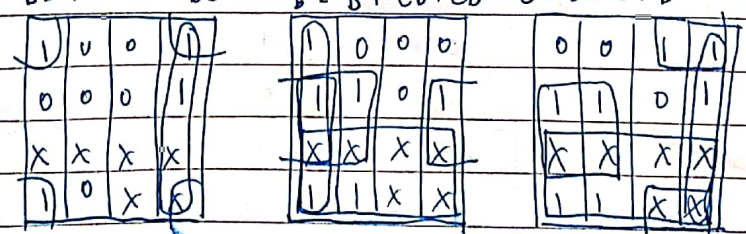
Q: Design a combinational circuit which takes input in BCD and converts into 7 segment display of equivalent BCD value.



A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	0	0	1	1



$a = \bar{B}\bar{D} + C + A + BD$ $b = \bar{B} + \bar{C}\bar{D} + CD$ $c = B + \bar{C} + D$ $d = A + C\bar{D} + \bar{B}\bar{D} + \bar{B}C + B\bar{C}D$



$e = \bar{B}\bar{D} + C\bar{D}$ $f = A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$ $g = A + C\bar{D} + \bar{B}C + B\bar{C}$

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Q: Design a combinational circuit ABCD that will multiply two 2-bit numbers

A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	0	0	1
0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	0	0	0	0	0
0	1	1	1	0	0	0	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0

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Comparator

x	y	a: x>y	b: x=y	c: x<y
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

	$\bar{y}$	y
$\bar{x}$	0	0
x	1	0

	$\bar{y}$	y
$\bar{x}$	1	0
x	0	1

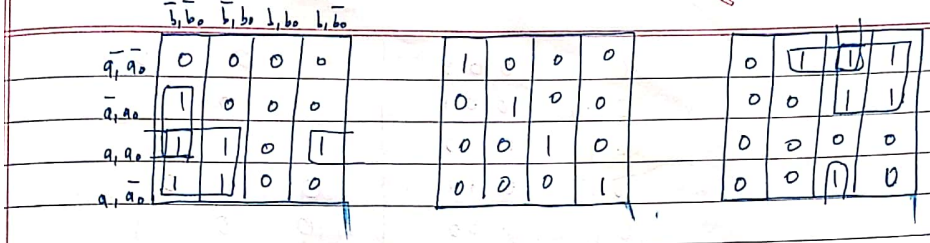
	$\bar{y}$	y
$\bar{x}$	0	1
x	0	0

$a = x\bar{y}$ $\quad b = \bar{x}\bar{y} + xy$ $\quad c = \bar{x}y$

$a_1 a_0$	$b_1 b_0$	x>y: a	x=y: b	x<y: c
0 0	0 0	0	1	0
0 0	0 1	0	0	1
0 0	1 0	0	0	1
0 0	1 1	0	0	1
0 1	0 0	1	0	0
0 1	0 1	0	1	0
0 1	1 0	0	0	1
0 1	1 1	0	0	1
1 0	0 0	1	0	0
1 0	0 1	1	0	0
1 0	1 0	0	1	0
1 0	1 1	0	0	1
1 1	0 0	1	0	0
1 1	0 1	1	0	0
1 1	1 0	1	0	0
1 1	1 1	0	1	0

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classmate
Date
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$a = a_1\bar{b}_1 + a_1a_0\bar{b}_0 + a_0\bar{b}_1\bar{b}_0$

$b = \bar{a}_1\bar{a}_0\bar{b}_1\bar{b}_0 + \bar{a}_1a_0\bar{b}_1b_0 + a_1a_0\bar{b}_1b_0 + a_1\bar{a}_0b_1\bar{b}_0$

$c = \bar{a}_1b_1 + \bar{a}_1\bar{a}_0b_0 + \bar{a}_0b_1b_0$

Q: Design and implement 3-bit comparator.

a_2 a_1 a_0	b_2 b_1 b_0	x>y x	x=y y	x<y z
0 0 0	0 0 0	0	1	0
0 0 0	0 0 1	0	0	1
0 0 0	0 1 0	0	0	1
0 0 0	0 1 1	0	0	1
0 0 0	1 0 0	0	0	1
0 0 0	1 0 1	0	0	1
0 0 0	1 1 0	0	0	1
0 0 0	1 1 1	0	0	1
0 0 1	0 0 0	1	0	0
0 0 1	0 0 1	0	1	0
0 0 1	0 1 0	0	0	1
0 0 1	0 1 1	0	0	1
0 0 1	1 0 0	0	0	1
0 0 1	1 0 1	0	0	1
0 0 1	1 1 0	0	0	1
0 0 1	1 1 1	0	0	1
0 1 0	0 0 0	1	0	0
0 1 0	0 0 1	1	0	0
0 1 0	0 1 0	0	1	0
0 1 0	0 1 1	0	0	1
0 1 0	1 0 0	0	0	1
0 1 0	1 0 1	0	0	1
0 1 0	1 1 0	0	0	1
0 1 0	1 1 1	0	0	1

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4 bit comparator

Greater than (X)

If $(A_3 > B_3)$ or (If $A_3 = B_3$ and $A_2 > B_2$) or (If $A_3 = B_3$, $A_2 = B_2$ and $A_1 > B_1$) or (if $A_3 = B_3$, $A_2 = B_2$, $A_1 = B_1$ and $A_0 > B_0$)

$$\Rightarrow A_3\bar{B_3} + X_3 A_2\bar{B_2} + X_3X_2 A_1\bar{B_1} + X_3X_2X_1 A_0\bar{B_0}$$

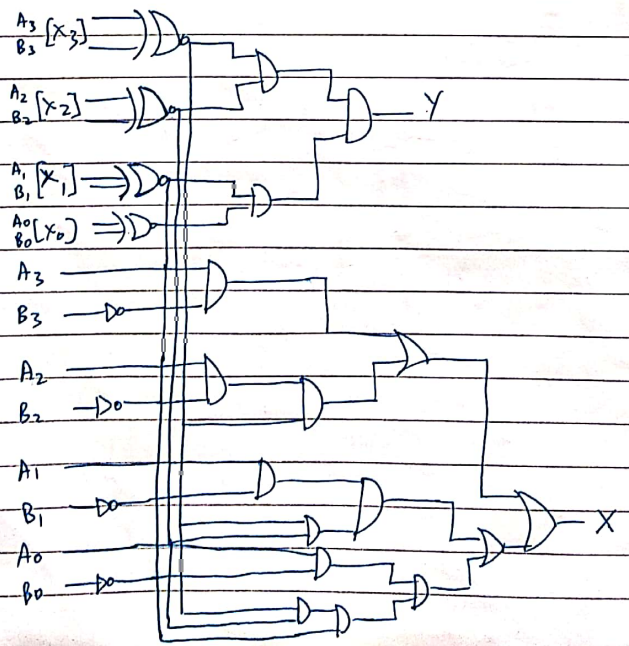
Lesser than (Z)

If $(A_3 < B_3)$ or if $(A_3 = B_3$ and $A_2 < B_2)$ or if $(A_3 = B_3, A_2 = B_2$ and $A_1 < B_1)$ or if $(A_3 = B_3, A_2 = B_2, A_1 = B_1$ and $A_0 < B_0)$

$$\Rightarrow \bar{A_3}B_3 + X_3\bar{A_2}B_2 + X_3X_2\bar{A_1}B_1 + X_3X_2X_1\bar{A_0}B_0$$

Equal to (Y)

$$\Rightarrow X_3X_2X_1X_0$$

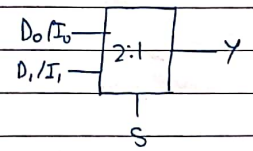


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Multiplexer (MUX)

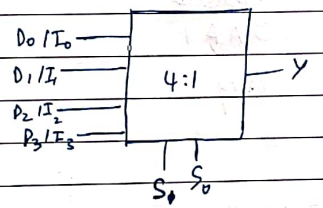
2:1 MUX



$Y = I_0 \bar{S} + I_1 S$

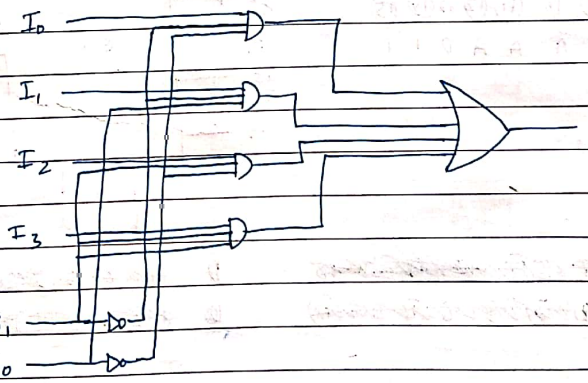
S	Y
0	I_0
1	I_1

4:1 MUX



S_1 S_0	Y
0 0	I_0
0 1	I_1
1 0	I_2
1 1	I_3

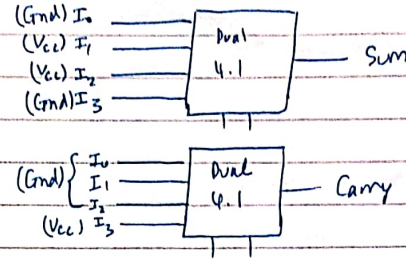
$Y = I_0 \bar{S_1}\bar{S_0} + I_1 \bar{S_1} S_0 + I_2 S_1 \bar{S_0} + I_3 S_1 S_0$



Half Adder using 4:1 MUX

Implementation table

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Q: Design full adder using dual 4:1 MUX.

$\bar{A}$	0	1	2	3
A	4	5	6	7
	A	$\bar{A}$	$\bar{A}$	A

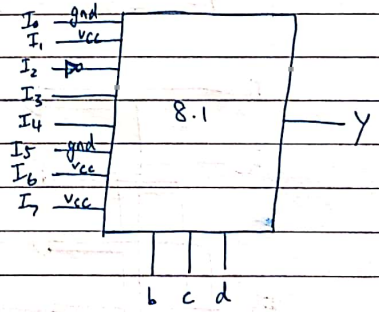
Sum

$\bar{A}$	0	1	2	3
A	4	5	6	7
	0	A	A	1

Carry

Q: $f(A,B,C,D) = \Sigma_m(1,2,6,7,9,11,12,14,15)$

$\bar{A}$	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	0	1	$\bar{A}$	A	A	0	1	1



Q: Implement 4:1 MUX using 2:1 MUX.

Q: Implement $u = ad + b\bar{c} + bd$ using (i) 16:1 MUX (ii) 8:1 MUX (iii) 4:1 MUX.

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Ans: $u = a(b+\bar{b})(c+\bar{c})d + (a+\bar{a})b\bar{c}(d+\bar{d}) + (a+\bar{a})b(c+\bar{c})d$

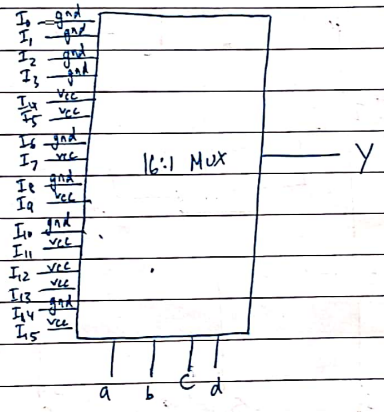
$= ab(c+\bar{c})d + a\bar{b}(c+\bar{c})d + ab\bar{c}(d+\bar{d}) + \bar{a}b\bar{c}(d+\bar{d})$
$+ ab(c+\bar{c})d + \bar{a}b(c+\bar{c})d$

$= abcd + ab\bar{c}d + a\bar{b}cd + a\bar{b}\bar{c}d + ab\bar{c}d + ab\bar{c}\bar{d}$
$+ \bar{a}b\bar{c}d + \bar{a}b\bar{c}\bar{d} + abcd + ab\bar{c}d + \bar{a}bcd + \bar{a}b\bar{c}d$

$= m_{15} + m_{13} + m_{11} + m_9 + m_{12} + m_5 + m_4 + m_7$

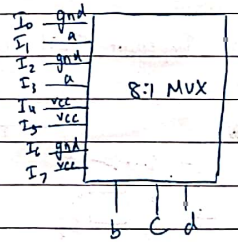
$= \Sigma_m(4,5,7,9,11,12,13,15)$

i)



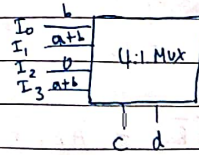
ii)

$\bar{a}$	0	1	2	3	4	5	6	7
a	8	9	10	11	12	13	14	15
	0	a	0	a	1	1	0	1



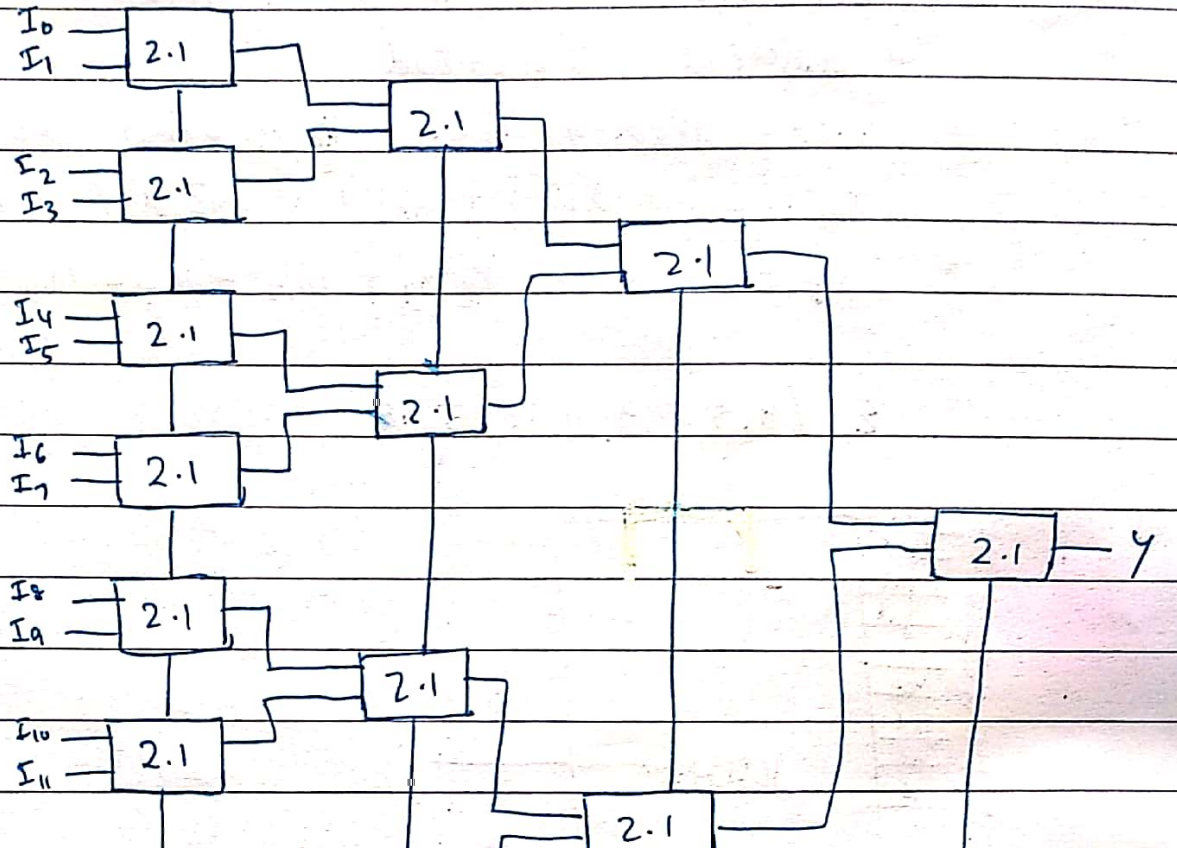
iii)

$\bar{a}\bar{b}$	0	1	2	3
$\bar{a}b$	4	5	6	7
ab	12	13	14	15
$a\bar{b}$	8	9	10	11
	b	a+b	0	a+b



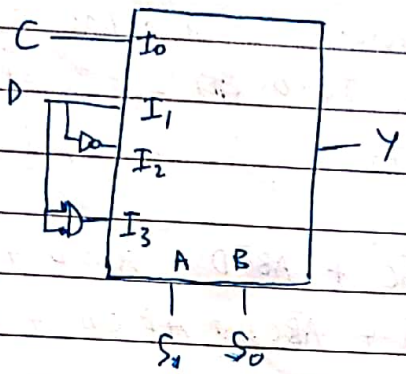
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Q: Implement 16:1 MUX using 2:1 MUX.



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Q:



Ans: $Y = \bar{A}\bar{B}C + \bar{A}BD + A\bar{B}\bar{D} + AB\bar{D}$

$= \bar{A}\bar{B}C(D+\bar{D}) + \bar{A}B(C+\bar{C})D + A\bar{B}(C+\bar{C})\bar{D} + AB(C+\bar{C})\bar{D}$

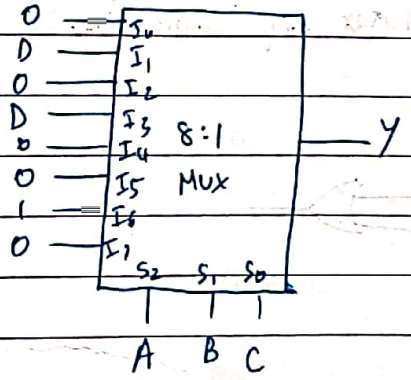
$= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}BCD + \bar{A}B\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D}$

$+ ABC\bar{D} + AB\bar{C}\bar{D}$

$= m_3 + m_2 + m_7 + m_5 + m_{10} + m_8 + m_{14} + m_{12}$

$= \Sigma_m(2, 3, 5, 7, 8, 10, 12, 14)$

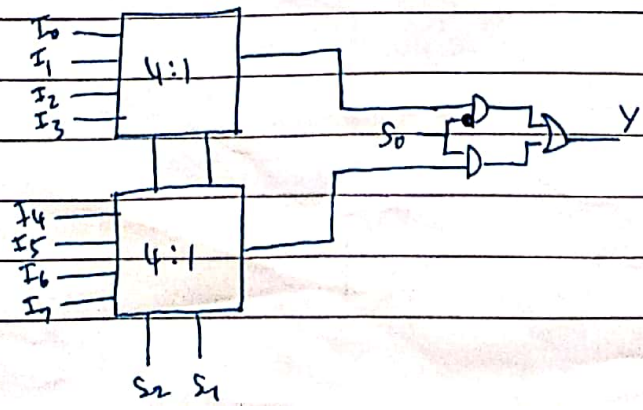
Q: 8:1 MUX is used to implement Y.



Ans: $Y = \bar{A}\bar{B}CD + \bar{A}BCD + AB\bar{C}\cdot 1$

$Y = \bar{A}CD + AB\bar{C}$

Q: Implement 8:1 MUX using only 4:1 MUX



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Q: 8:1 MUX has input a,b,c connected to selection line S_2, S_1, S_0. The inputs through $I_0 - I_7$ are (i) $I_1 = I_2 = I_4 = 0$ $I_3 = I_5 = 1$, $I_0 = I_7 = D$, $I_6 = \bar{D}$

(ii) $I_2 = I_3 = 0$ $I_4 = I_5 = I_7 = 1$ $I_0 = I_6 = D$ $I_1 = \bar{D}$

Ans: i) $Y = \bar{A}BC + A\bar{B}C + \bar{A}\bar{B}\bar{C}D + ABCD + AB\bar{C}\bar{D}$

$= \bar{A}BCD + \bar{A}BC\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} + ABCD + \bar{A}\bar{B}\bar{C}D + AB\bar{C}\bar{D}$

$= m_7 + m_6 + m_{11} + m_{10} + m_{15} + m_1 + m_{12}$

$= \Sigma m (1, 6, 7, 10, 11, 12, 15)$

ii) $Y = A\bar{B}\bar{C} + A\bar{B}C + ABC + \bar{A}\bar{B}\bar{C}D + AB\bar{C}D + \bar{A}\bar{B}CD$

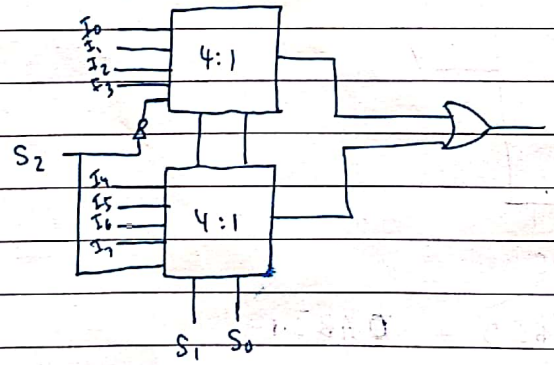
$= A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} + ABCD + ABC\bar{D} + \bar{A}\bar{B}CD$

$+ AB\bar{C}D + \bar{A}\bar{B}C\bar{D}$

$= m_9 + m_8 + m_{11} + m_{10} + m_{15} + m_{14} + m_1 + m_{13} + m_2$

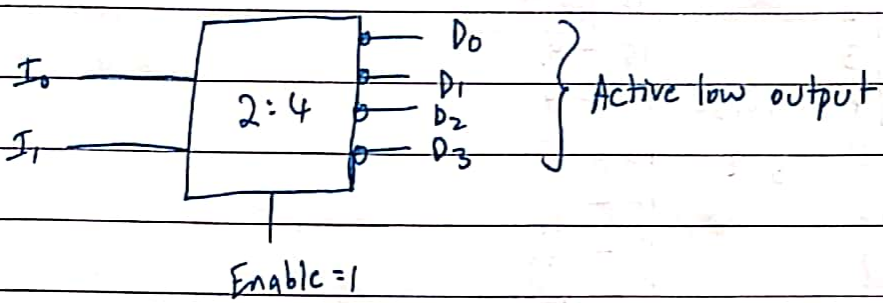
$= \Sigma m (1, 2, 8, 9, 10, 11, 13, 14, 15)$

Q: Implement 8:1 MUX using 4:1 MUX using enable line.



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Decoders / Demultiplexers

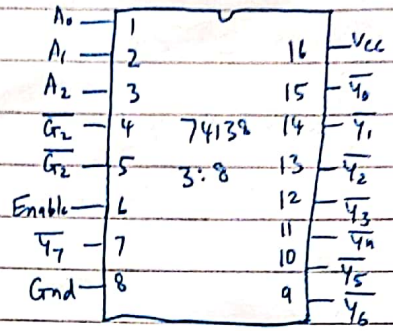


Enable	I_0	I_1	D_0	D_1	D_2	D_3
0	X	X	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

Q: Design and implement 3:8 decoder using logic gates.

Enable	I_0	I_1	I_2	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	X	X	X	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0

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Q: Full adder using 74138.

Q: m-bit thermometer code converter for k consists of k 1's in LSB and (m-k) 0's in MSB. A binary to thermometer converter has n inputs and $2^n - 1$ outputs. It produces $2^n - 1$ bit thermometer code for the number specified by input.

Eg: 110 → 0111111

A B C	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0 0 0	0	0	0	0	0	0	0
0 0 1	0	0	0	0	0	0	1
0 1 0	0	0	0	0	0	1	1
0 1 1	0	0	0	0	1	1	1
1 0 0	0	0	0	1	1	1	1
1 0 1	0	0	1	1	1	1	1
1 1 0	0	1	1	1	1	1	1
1 1 1	1	1	1	1	1	1	1

$Y_6 = ABC$ $Y_5 = AB$ $Y_3 = AB + A\bar{B}$

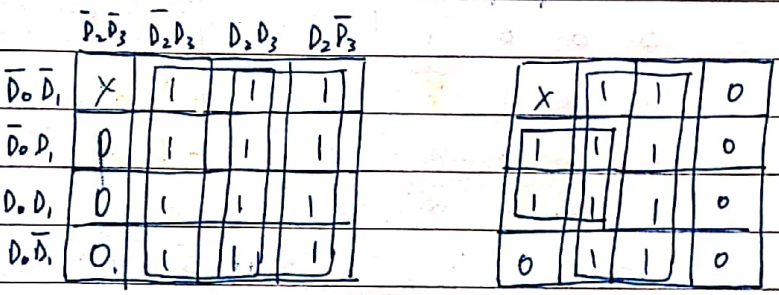
$Y_4 = AB + AC$ $= A$ $Y_2 = A + BC$

$Y_1 = A + B$ $Y_0 = A + B + C$

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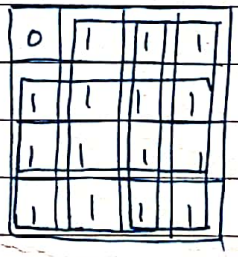
Encoders and Priority Encoder

D_0	D_1	D_2	D_3	X	Y	Validity
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1
0	0	0	0	X	X	0



$X = D_2 + D_3$

$Y = D_3 + D_1\bar{D}_2$



$V = D_1 + D_0 + D_2 + D_3$

Q: Design octal to binary encoder in which MSB will have more priority.

D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	X	Y	Z	V
1	0	0	0	0	0	0	0	0	0	0	1
X	1	0	0	0	0	0	0	0	0	1	1
X	X	1	0	0	0	0	0	0	1	0	1
X	X	X	1	0	0	0	0	0	1	1	1
X	X	X	X	1	0	0	0	1	0	0	1
X	X	X	X	X	1	0	0	1	0	1	1
X	X	X	X	X	X	1	0	1	1	0	1
X	X	X	X	X	X	X	1	1	1	1	1
0	0	0	0	0	0	0	0	X	X	X	0

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$X = D_4 + D_5 + D_6 + D_7$

$Y = D_2 + D_3 + D_6 + D_7$

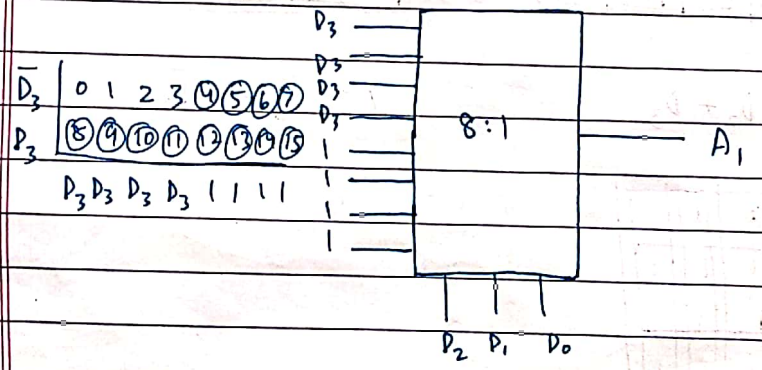
$Z = D_1 + D_3 + D_5 + D_7$

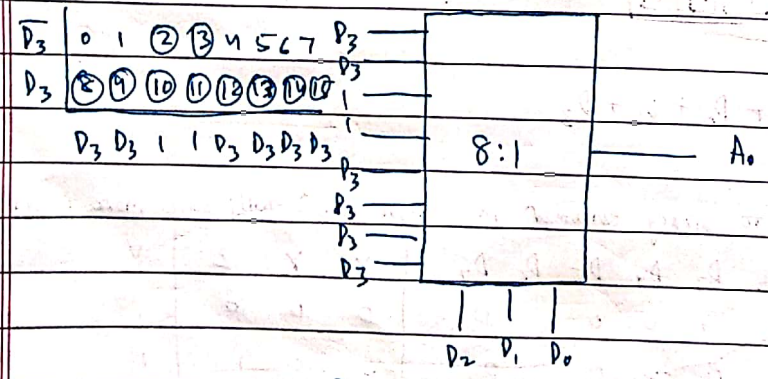
$V = D_0 + D_1 + D_2 + D_3 + D_4 + D_5 + D_6 + D_7$

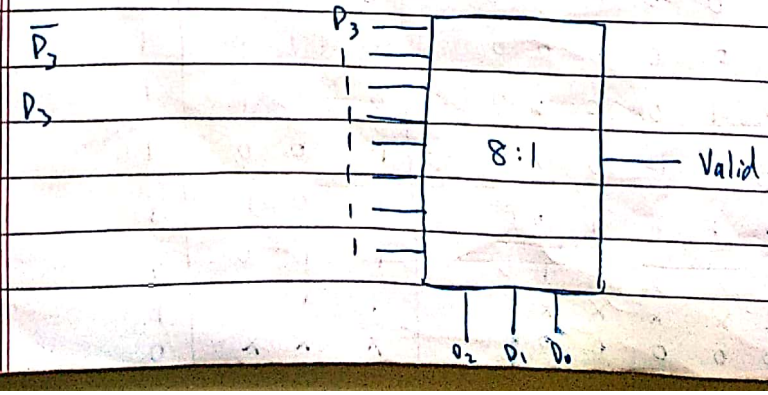
Q: Implement 4:2 priority encoder using 8:1 MUX.

Ans:

D_3	D_2	D_1	D_0	A_1	A_0	Valid
0	0	0	0	x	x	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1







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Sequential Circuits

* Combinational circuits depends only on present input; Sequential circuit depends on present input and past output, i.e, it stores data/ has memory (flip-flop/latch)
* Combinational circuit does not have memory and no feedback path; Sequential circuit has memory and feedback path
* Combinational circuit is easy to design and is faster; Sequential circuit is harder to design and is slower

Latch vs Flip flop

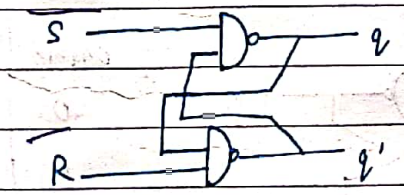
* Latch does not consider clock; Flip-flop considers clock
* Latches are asynchronous ; Flip-flops are synchronous
* Latches are level triggered; Flip-flops are edge triggered

Types of Latches

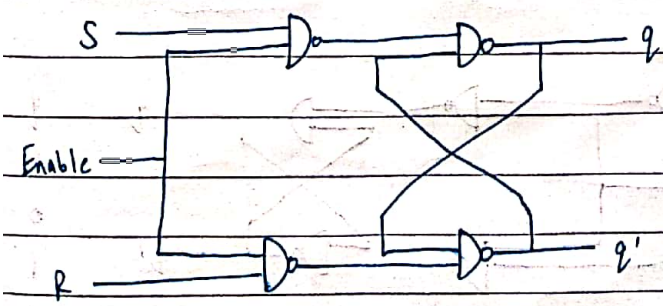
* S R (Set Reset) latch
* D latch (Delay / Data)
* J K latch (Jack Kill)
* T (Toggle) latch

} When they are connected to clock, they become flip flop.

S R latch



S	R	q	q'
0	0	1	1
0	1	1	0
1	0	0	1
1	1	0	0 (Race condition)



Enable	S	R	q q'
1	0	0	No change
1	0	1	Reset
1	1	0	Set
1	1	1	Invalid

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Using clock, $q(t)$ instead of enable line

$q(t)$	S	R	$q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	Invalid
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	Invalid

	$\bar{S}\bar{R}$	$\bar{S}R$	SR	$S\bar{R}$
$\bar{q}$	0 (0)	0 (1)	X (3)	1 (2)
q	1 (4)	0 (5)	X (7)	1 (6)

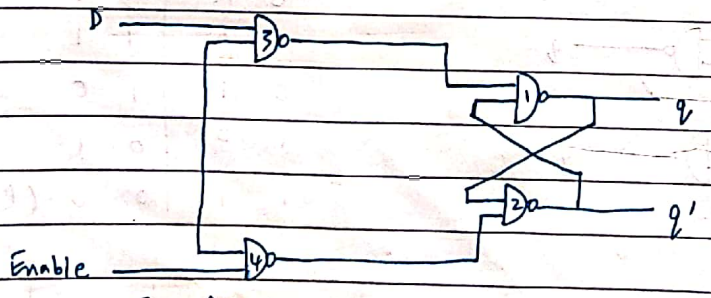
$$S + q\bar{R}$$

$$q(t+1) = S + q\bar{R}$$

Excitation table

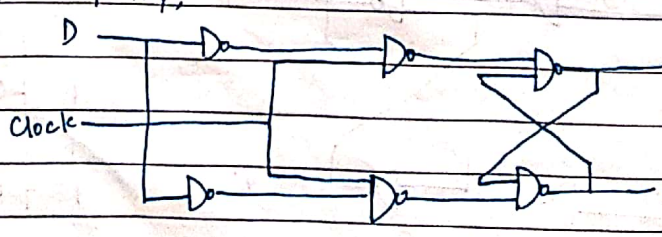
$q(t)$	$q(t+1)$	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

D latch



Enable	D	$q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

For flip-flop,

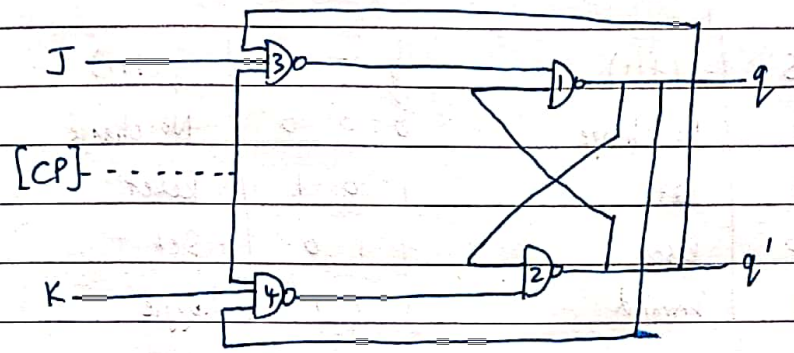


q	D	$q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

$q(t+1) = D$

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J K latch



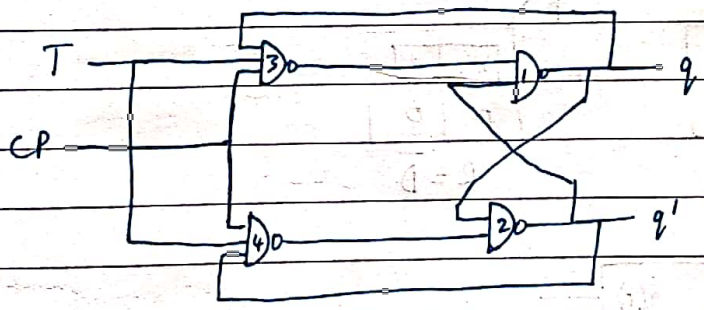
Flip-flop

q	J	K	q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

For flip-flop, connect gates 3 & 4 to clock pulse

$$q(t+1) = Jq' + K'q$$

For Toggle flip-flop,



q	T	q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

$$q(t+1) = T \oplus q$$

Excitation tables

q(t)	q(t+1)	D
0	0	0
0	1	1
1	0	0
1	1	1

q(t)	q(t+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

q(t)	q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

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Characteristic Table

q	R	S	q(t+1)
0	0	0	No change
1	0	1	Set
1	1	0	Reset
1	1	1	Invalid

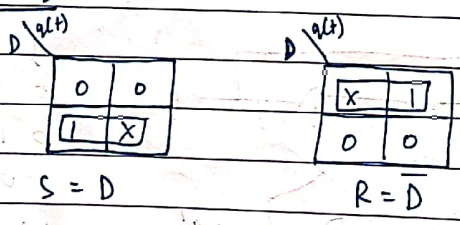
q	J	K	q(t+1)
0	0	0	No change
1	0	1	Reset
1	1	0	Set
1	1	1	Toggle

q	D	q(t+1)
0	0	No change
0	1	Set

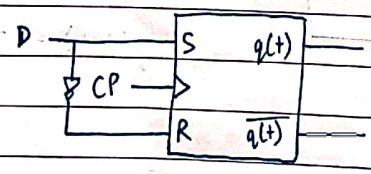
q	T	q(t+1)
0	0	No change
0	1	Set

Conversion of flip-flops

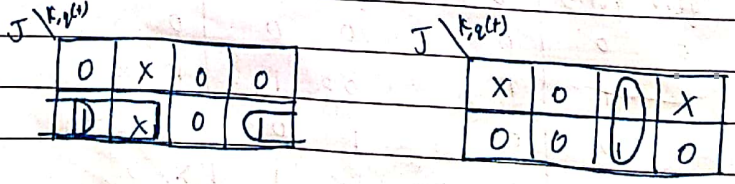
* **SR to D**



$S = D$ $\quad R = \overline{D}$



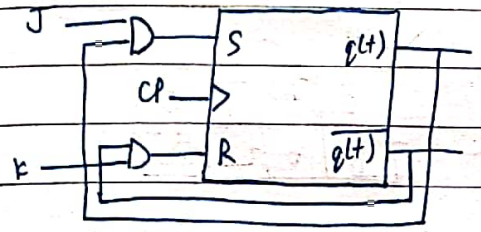
* **SR to JK**



$S = J\overline{q(t)}$ $\quad R = Kq(t)$

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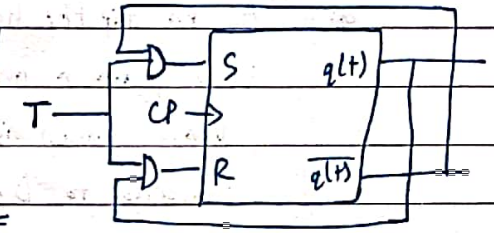


* SR to T

T \ Q(t)	0	1
0	X	0
1	0	1

$S = T Q^{(t)}$

$R = T \overline{Q^{(t)}}$

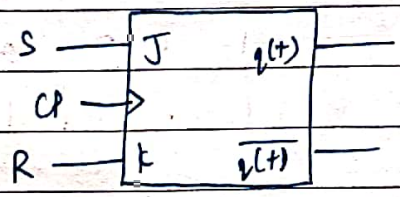


* JK to SR

S \ R, Q(t)	0,0	0,1	1,0	1,1
0	X	X	0	0
1	1	X	X	X

$J = S$

$K = R$

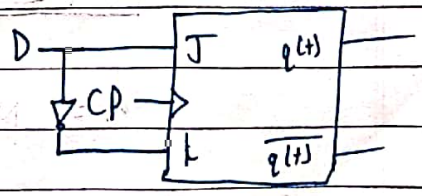


* JK to D

D \ Q(t)	0	1
0	0	X
1	1	X

$J = D$

$K = \overline{D}$



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* D to SR

S \ R, q(t)	0	1	0	0
	1	1	X	X

$D = S + \bar{R} q(t)$

* T to SR

S \ R, q(t)	0	0	1	0
	1	0	X	X

$T = R q(t) + S \overline{q(t)}$

Q: A pn flip flop has 4 operations. Clear to 0, No change, Complement, Set to 1. When p and n are 00, 01, 10, 11 respectively. Tabulate the characteristic table. Derive characteristic equation. Tabulate excitation table. Show how pn can be converted to D.

Ans:

Characteristic Table

P	N	q(t+1)
0	0	0
0	1	q(t)
1	0	$\overline{q(t)}$
1	1	1

q(t)	P	N	q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

q(t) \ PN	0	0	1	1
	0	1	1	0

$q(t+1) = N q(t) + \overline{q(t)} P$

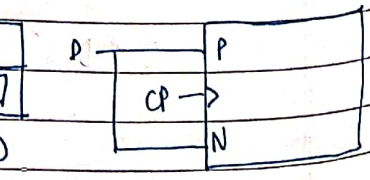
D	q(t)	q(t+1)	P	N
0	0	0	0	X
1	0	1	1	X
0	1	0	X	0
1	1	1	X	1

D \ q(t)	0	X
	1	X

P = D

D \ q(t)	X	0
	X	1

N = D



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Q: Show that the characteristic equation for the complement output of JK ff is

$$\overline{q(t+1)} = \bar{J}\bar{q} + \bar{K}q$$

Ans:

J	K	$\overline{q(t+1)}$
0	0	$\overline{q(t)}$
0	1	1
1	0	0
1	1	$q(t)$

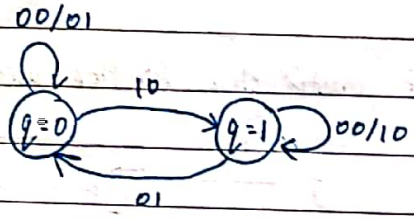
q(t)	J	K	$\overline{q(t+1)}$
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

q(t) \ JK	11	10	00	01
1	1	0	0	
0		1	1	0

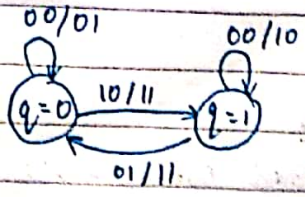
$$\overline{q(t+1)} = q(t)\cdot K + \overline{q(t)}\cdot\bar{J}$$

Q: Represent the state transition diagram for all types of flip-flops.

q	S	R	q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

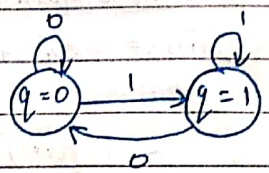


q	J	K	q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

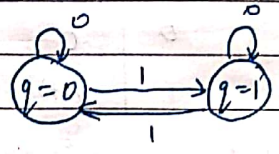


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q	D	q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1



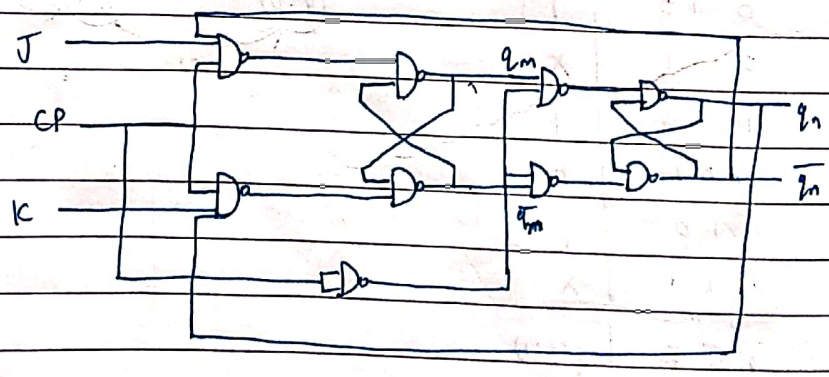
q	T	q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0



* Truth table describes the combinational circuits.
* State table describes the sequential circuits.

* Characteristic table describes the operation of flip-flop.
* Excitation table is used to identify the flip-flop input by using present state and next state.

Q: Construct master slave flip flop using JK along with timing diagram.



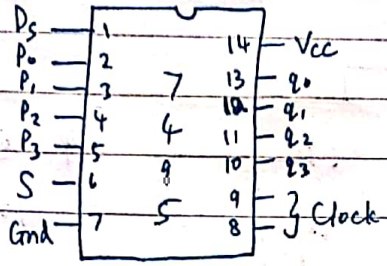
Registers

Shift Registers

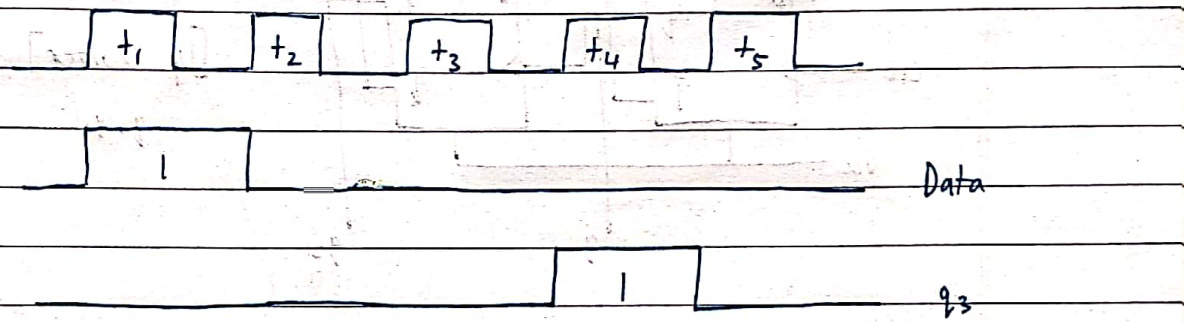
Serial In Serial Out (SISO)
Serial In Parallel Out (SIPO)
Parallel In Serial Out (PISO)
Parallel In Parallel Out (PIPO)

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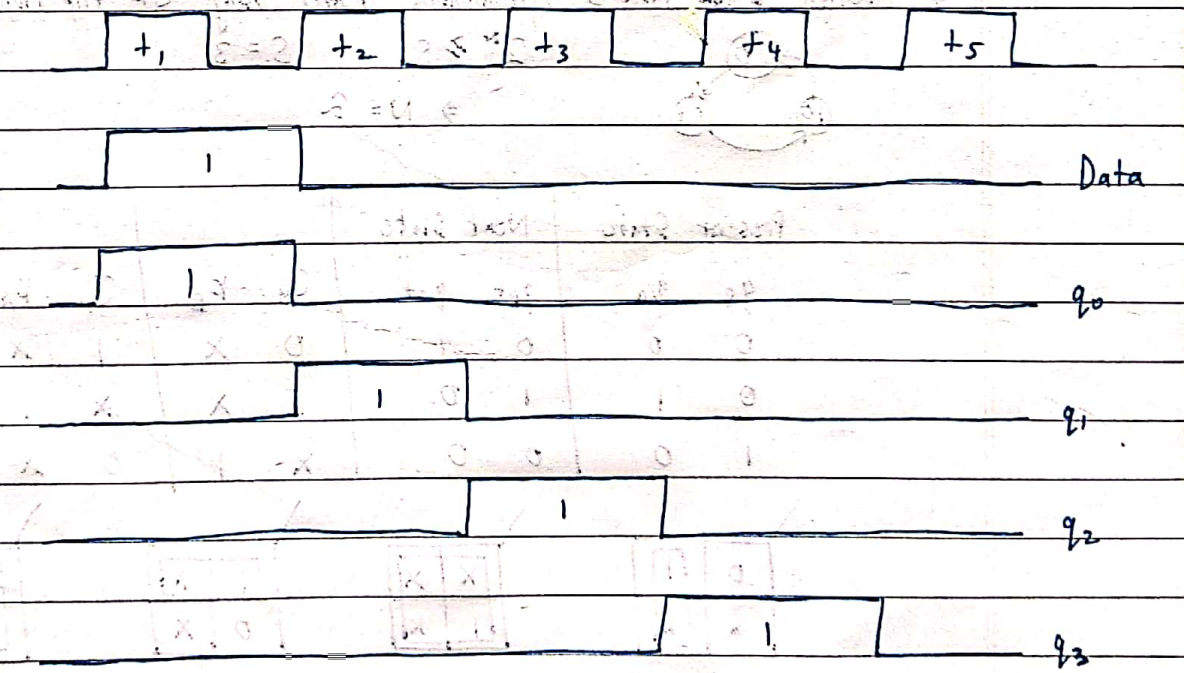
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* **SISO:**



* **SIPO:**



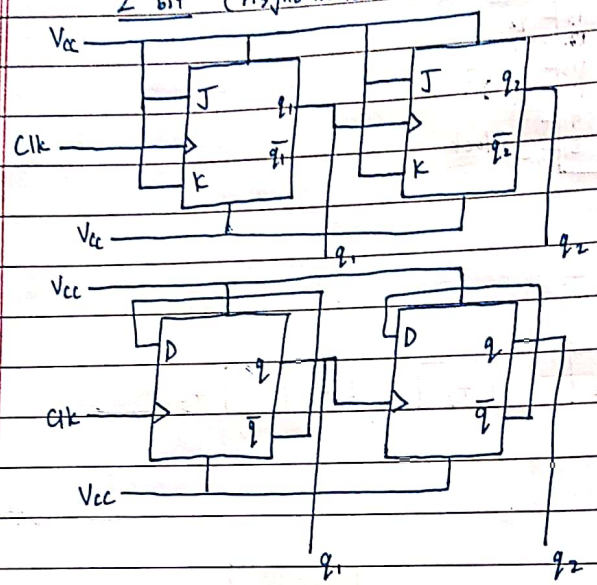
Design of Counters

* Synchronous → Clock pulse applied to all flip-flops

* Asynchronous → Clock pulse applied to only first flip-flop, output of previous flip flop is clock for next.

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2 bit (Asynchronous)



[Up Counter]

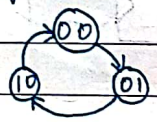
$q_2 q_1$

[Up Counter]

$q_2 q_1$

Also called as ripple counter

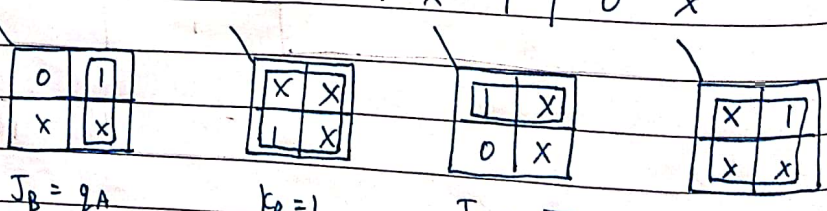
Q: Design & mod 3 synchronous counter using JK flip flop.



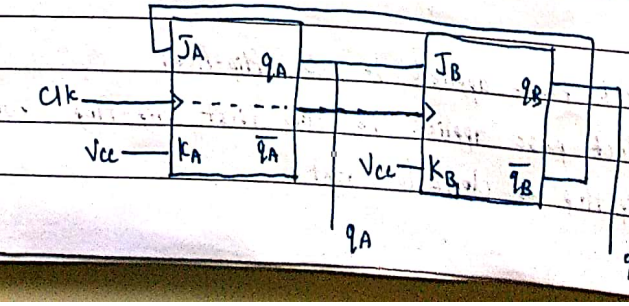
$2^N \geq S \qquad S = 3$

$\Rightarrow N = 2$

Present State		Next State					
q_B	q_A	q_{B+}	q_{A+}	J_B	K_B	J_A	K_A
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	0	0	X	1	0	X

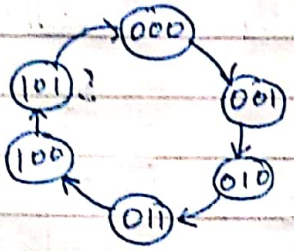


$J_B = q_A$ $\qquad K_B = 1$ $\qquad J_A = \bar{q}_B$ $\qquad K_A = 1$



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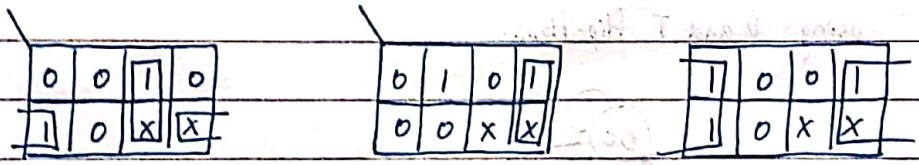
Q: Design mod 6 synchronous counter using D flip flop.



$2^N \geq S \quad S = 6$

$\Rightarrow N = 3$

q_C	q_B	q_A	q_C^+	q_B^+	q_A^+	D_C	D_B	D_A
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	0	1	1	0	1	1
0	1	1	1	0	0	1	0	0
1	0	0	1	0	1	1	0	1
1	0	1	0	0	0	0	0	0

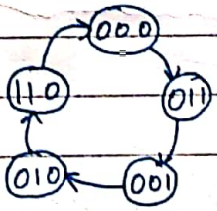


$D_C = q_B q_A + q_C \overline{q_A}$

$D_B = \overline{q_C}\,\overline{q_B} q_A + q_B \overline{q_A}$

$D_A = \overline{q_A}$

Q: Design synchronous counter using SR flip flop that generates $0 \to 3 \to 1 \to 2 \to 6$



$2^N \geq S \quad S = 5$

$\Rightarrow N = 3$

q_C	q_B	q_A	q_C^+	q_B^+	q_A^+	S_C	R_C	S_B	R_B	S_A	R_A
0	0	0	0	1	1	0	X	1	0	1	0
0	1	1	0	0	1	0	X	0	1	X	0
0	0	1	0	1	0	0	X	1	0	0	1
0	1	0	1	1	0	1	0	X	0	0	X
1	1	0	0	0	0	0	1	0	1	0	X

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0	0	0	1
X	X	X	0

$S_C = \bar{q}_C q_B \bar{q}_A$

X	X	X	0
X	X	X	1

$R_C = q_C$

1	1	0	X
X	X	X	0

$S_B = \bar{q}_B$

0	0	1	0
X	X	X	1

$R_B = q_C + q_B q_A$

1	0	X	0
X	X	X	0

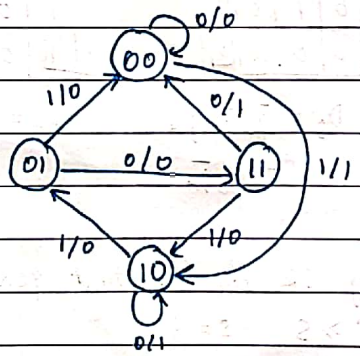
$S_A = \overline{q_B \oplus q_A}$

0	1	0	X
X	X	X	X

$R_A = q_B \oplus q_A$

Analysis of Flip-Flops

Q: Sequential circuit has 1 input and 1 output. State diagram is shown. Design circuit using D and T flip-flops.



State Table

Present State	Input	Next State	T_A	T_B	Y
00	0	00	0	0	0
00	1	10	1	0	1
01	0	11	1	0	0
01	1	00	0	1	0
10	0	10	0	0	1
10	1	01	1	1	0
11	0	00	1	1	1
11	1	10	0	1	0

Q:

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	$\bar{B}\bar{x}$	$\bar{B}x$	Bx	$B\bar{x}$
$\bar{A}$	0	1	0	1
A	0	1	0	1

0	0	1	0
0	1	1	1

$T_A = \bar{B}x + B\bar{x}$

$= B \oplus x$

$T_B = AB + Bx + Ax$

0	1	0	0
1	0	0	1

$Y = A\bar{x} + \bar{A}\bar{B}x$

Present State	Input	Next state	D_A	D_B	Y
00	0	00	0	0	0
00	1	10	1	0	1
01	0	11	1	1	0
01	1	00	0	0	0
10	0	10	1	0	1
10	1	01	0	1	0
11	0	00	0	0	1
11	1	10	1	0	0

0	1	0	1
0	0	1	0

0	0	0	1
0	1	0	0

0	1	0	0
1	0	0	1

$D_A = A \oplus B \oplus x$

$D_B = A\bar{B}x + \bar{A}B\bar{x}$

$Y = A\bar{x} + \bar{A}\bar{B}x$

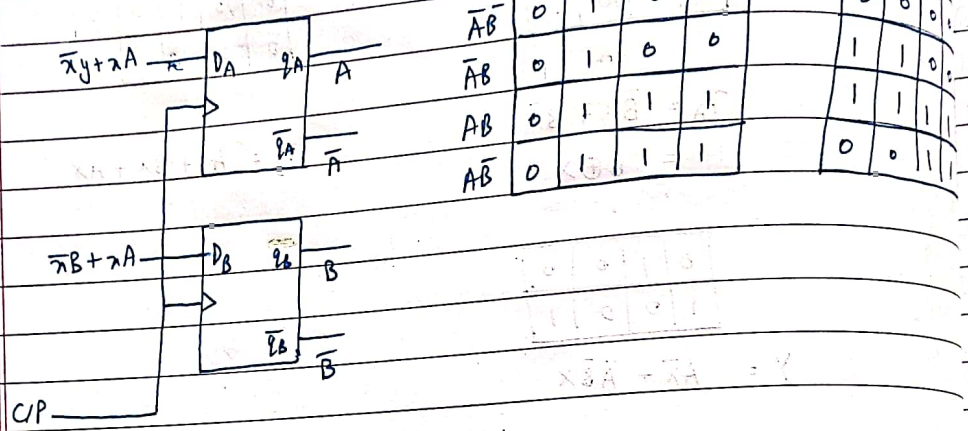
2 D flip flops A & B, 2 inputs X & Y, 1 output Z is specified by next-state output equation, $A(t+1) = x'y + xA$

$B(t+1) = x'B + xA$

$Z = B$

Draw logic diagram of circuit. List the state table for sequential circuit. Draw the corresponding state diagram.

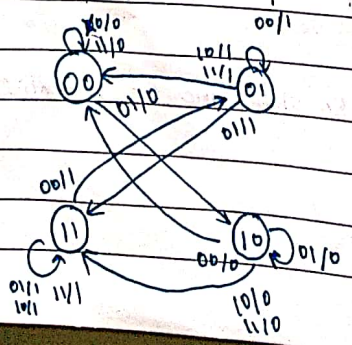
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	$\bar{x}\bar{y}$	$\bar{x}y$	xy	$x\bar{y}$
$\bar{A}\bar{B}$	0	1	0	0
$\bar{A}B$	0	1	0	0
AB	0	1	1	1
$A\bar{B}$	0	1	1	1

0	0	0
1	1	0
1	1	1
0	0	1

Present State		Input		Next State		
A	B	X	Y	A^+	B^+	Z
0	0	0	0	0	0	0
0	0	0	1	1	0	0
0	0	1	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	0	0
1	0	0	1	1	0	0
1	0	1	0	1	1	0
1	0	1	1	1	1	0
1	1	0	0	0	1	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1



Q: Design circuit
state of
01 →

A
0
0
0
0
1
1
1
1

$\bar{A}$
A

D_A

Moore

M

* If the
state
* Input
* More

Q: Design

Ans:

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Design circuit with 2 D flip flops A & B and 1 input X. When x=0, the state of circuit remains the same. x=1, circuit goes through from 00 → 01 01 → 11, 11 → 10, 10 → 00.

A	B	X	A^+	B^+
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	1
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	1	0

	$\bar{B}\bar{X}$	$\bar{B}X$	BX	$B\bar{X}$
$\bar{A}$	0	0	1	0
A	1	0	1	1

$$D_A = BX + A\bar{X}$$

	$\bar{B}\bar{X}$	$\bar{B}X$	BX	$B\bar{X}$
$\bar{B}$	0	1	1	1
B	0	0	0	1

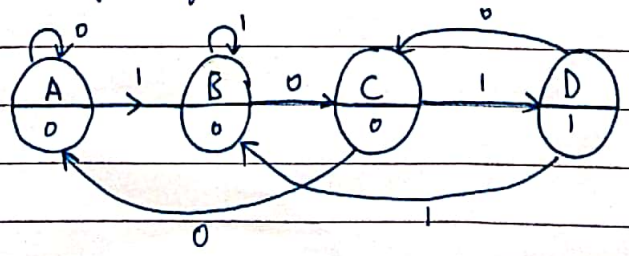
$$D_B = B\bar{X} + \bar{A}X$$

Moore & Mealy Model

Moore Model	Mealy Machine
* If the output depends on present state	* If the output depends on present state and present input
* Input changes do not affect output	* Input changes affect the output
* More number of states (n+1)	* Less number of states (n)

Q: Design Moore type sequence detector to detect 101.

Ans:



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PS	NS X=0	NS X=1	O/P
A	A	B	0
B	C	B	0
C	A	D	0
D	C	B	1

PS A	PS B	NS X=0	NS X=1	O/P
0	0	0 0	0 1	0
0	1	1 0	0 1	0
1	0	0 0	1 1	0
1	1	1 0	0 1	1

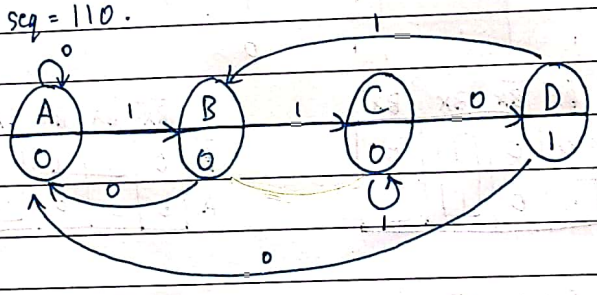
X=0			
D_A	D_B	D_A	
0	0	0	
1	0	0	
0	0	1	
1	0	0	

$D_A = A\bar{B}x + B\bar{x}$

$D_B = X$

$O/P = AB$

Q: Design a sequence detector to detect sequence using JK, considering Moore machine, seq = 110.



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